Construction Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

Construction project planning is a complex process involving multiple tasks, resource constraints, time and cost estimations, and risk factors. Manual planning methods are often inefficient, error-prone, and lack flexibility when handling dependencies and dynamic changes.

There is a need for an intelligent system that can **accept a high-level construction goal and autonomously generate a structured, optimized execution plan** while validating resources, estimating costs and time, and visualizing the schedule clearly.

1.2 Project Objectives

The objectives of the Construction Planning Assistant Agent are:

- To design an intelligent agent capable of autonomous construction planning
- To decompose high-level construction goals into actionable tasks
- To validate availability of required resources
- To estimate time and cost for construction activities
- To optimize task execution order
- To analyze risks related to delays and dependencies
- To present the final plan through a clear and interactive web interface

1.3 Scope of the Project

- Academic-level construction planning assistant
- Single goal-based planning per execution
- In-memory data handling without persistent storage
- Mock resource validation for demonstration purposes
- Stateless backend execution

1.4 Out of Scope:

- Real-time construction monitoring

- Integration with live industry databases
- Automatic execution-time re-planning

2. Proposed Solution

2.1 Key Features

- AI-based task decomposition
- Resource availability validation
- Time and cost estimation
- Optimized task scheduling
- Risk analysis
- Gantt chart visualization
- Web-based user interface

2.2 Overall Architecture / Workflow

2.2.1 High Level Design Summary

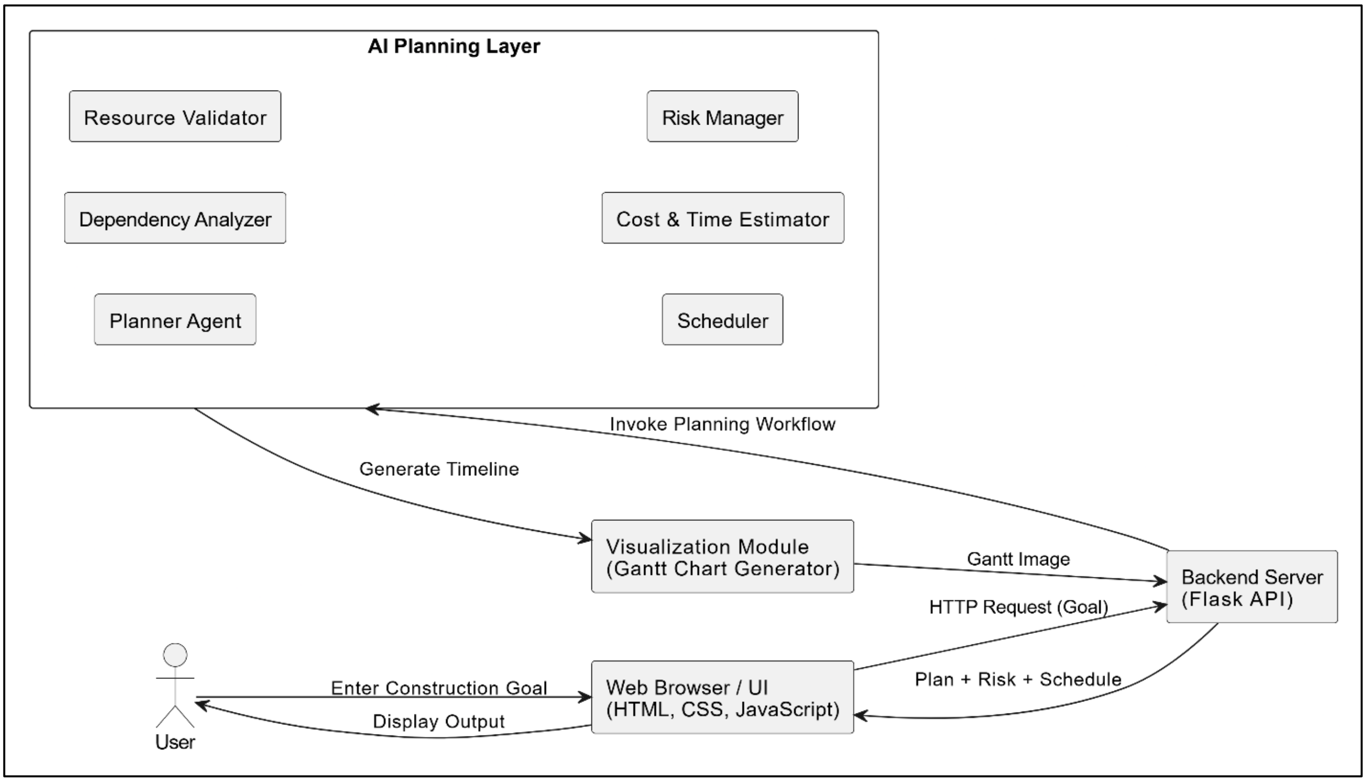

Fig. High Level Architecture Block Diagram

The Construction Planning Assistant Agent is designed using a modular, agent-based approach. Each agent is responsible for a specific planning function such as task decomposition, resource validation, scheduling, and risk analysis.

The backend operates in a **stateless manner**, processing each request independently using in-memory data structures. The system prioritizes simplicity, clarity, and scalability, making it suitable for academic evaluation and future enhancements.

2.2.2 Workflow Description

1. The user enters a high-level construction goal through the web interface
2. The goal is sent to the backend using a REST API
3. The Planner Agent decomposes the goal into smaller tasks
4. Resource availability is validated
5. Time and cost estimates are calculated
6. Task execution order is optimized
7. Risks related to delays are evaluated
8. A Gantt chart is generated
9. The complete optimized plan is displayed on the frontend

This workflow reflects the **High-Level Design (HLD)** of the system and ensures clarity, modularity, and traceability.

2.2.3 Information Flow

- Input: User Goal
- Processing: Task data, dependency graph, estimates
- Output: Optimized schedule, risks, visual charts

Data flows **unidirectionally** from UI → Backend → UI ensuring clarity and traceability.

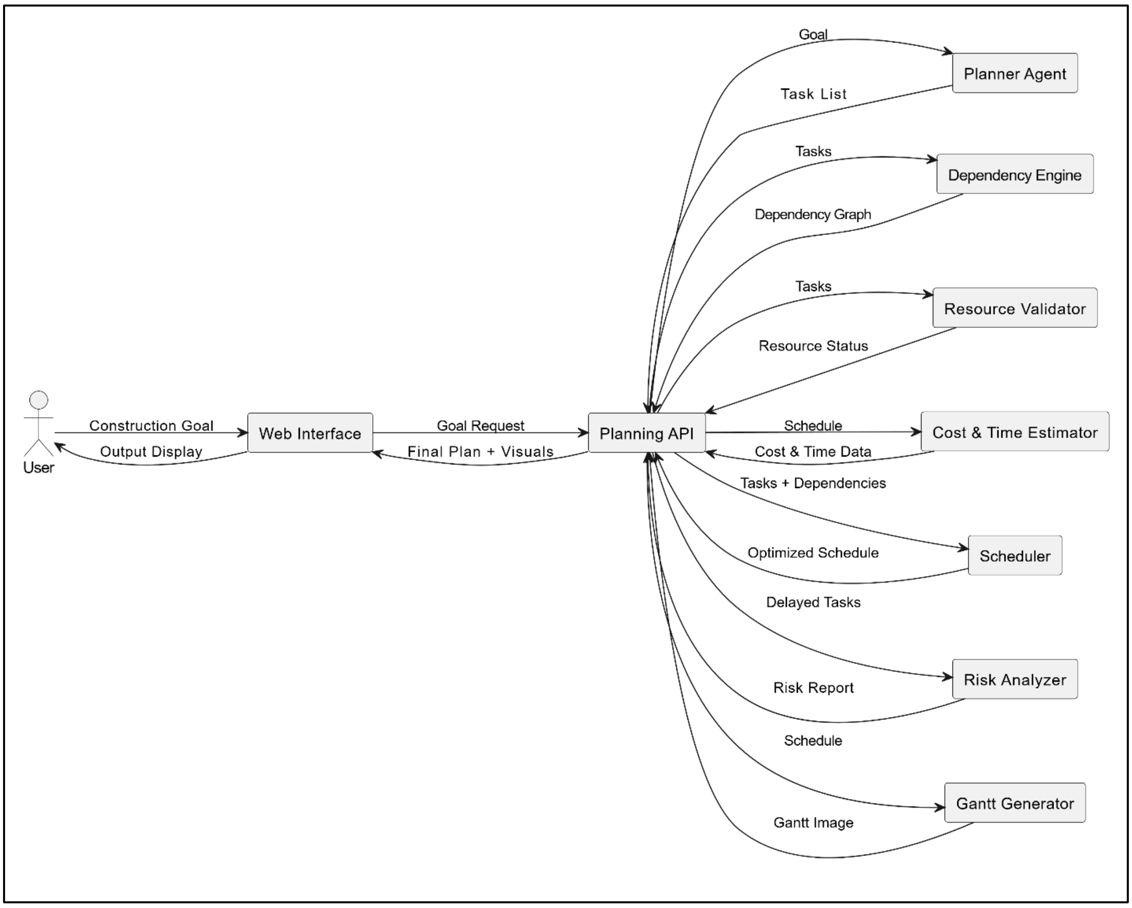

Fig. Dataflow Diagram

2.3 Tools & Technologies Used

Frontend

- HTML
- CSS
- JavaScript

Backend

- Python
- Flask

AI & Planning

- Groq LLM (for intelligent task decomposition)
- Rule-based fallback planning logic

Visualization

- Gantt chart generation using Python

3. Results & Output

3.1 User Interface:

Below Figure shows the web-based interface of the Construction Planning Assistant Agent. The interface allows users to input high-level construction goals and displays the generated plan, risk analysis, and Gantt chart visualization.

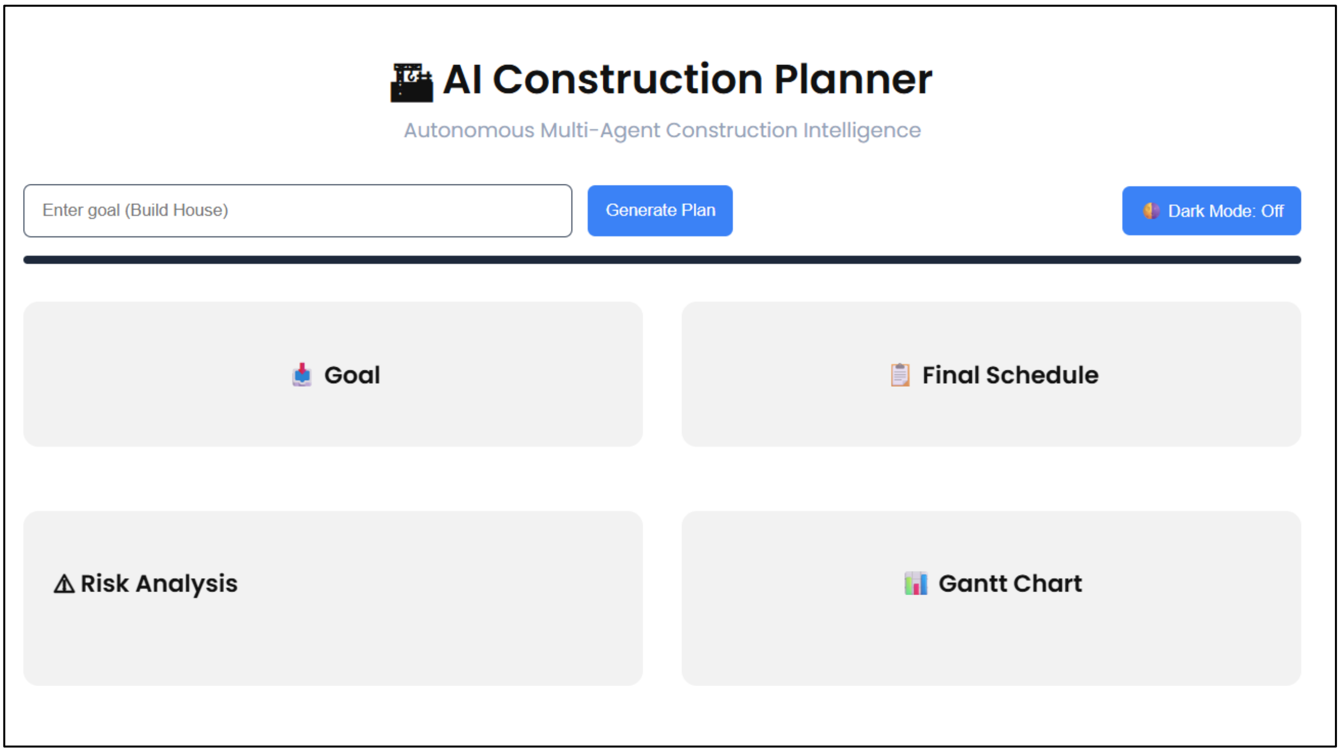

Fig. Web Interface of the Construction Planning Assistant Agent

Below figure shows the complete output generated by the Construction Planning Assistant Agent after processing a high-level construction goal, including the final task schedule, identified risks, and Gantt chart visualization.

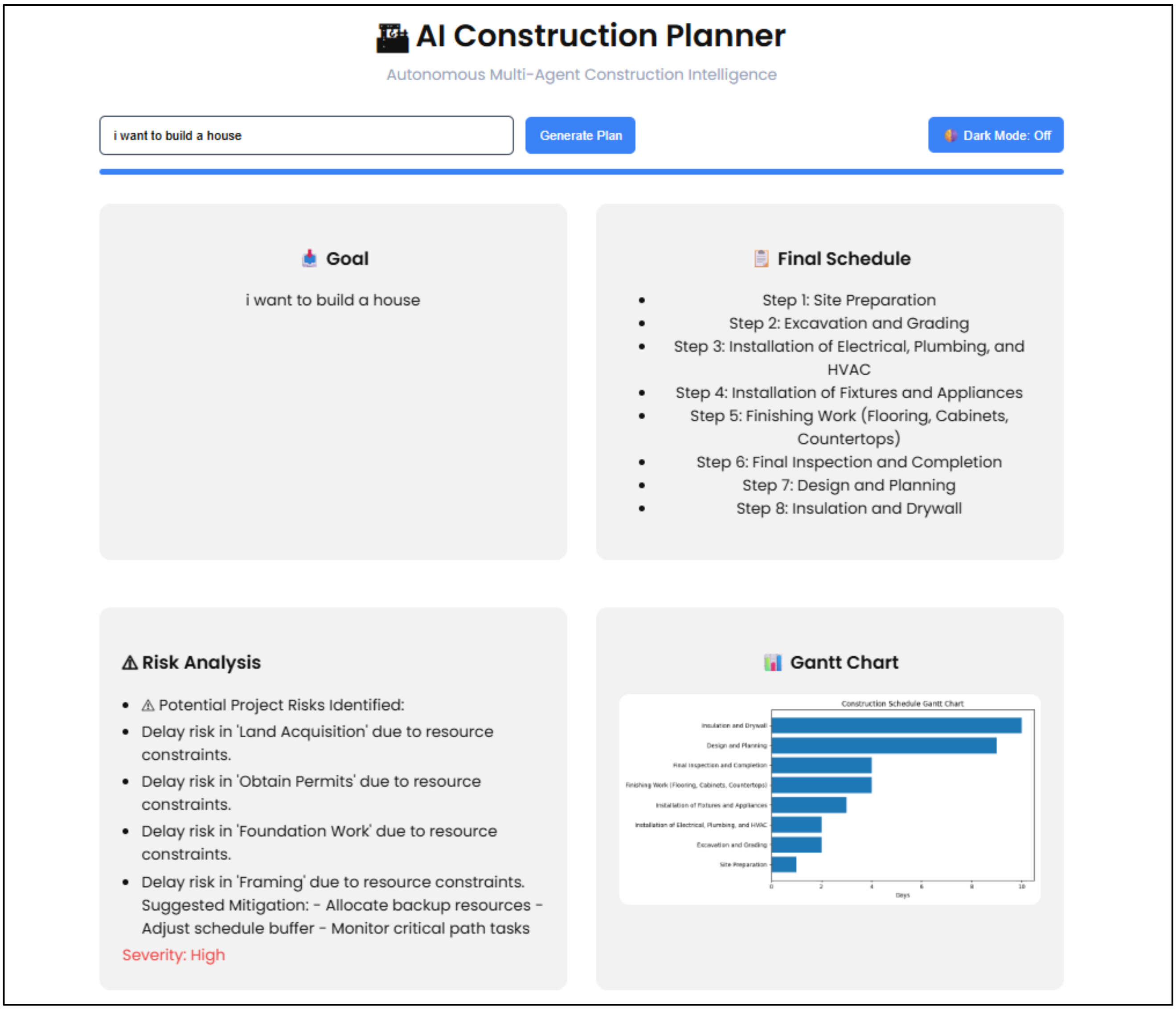

Fig. Generated Construction Plan with Risk Analysis and Gantt Chart

3.2 Reports / Dashboards / Models

- **Construction Execution Plan:**

  The system generates a structured execution plan by decomposing the high-level construction goal into sequential tasks. The plan outlines the order of execution and provides a clear roadmap for completing the construction project.

- **Time and Cost Estimation Report:**

  Based on the identified tasks and available resources, the system estimates the approximate time and cost required for each construction activity. These estimates help in understanding project feasibility and planning efficiency.

- **Risk Analysis Output:**

  The agent identifies potential risks such as delays due to resource constraints and task dependencies. It also suggests mitigation strategies like resource reallocation and schedule buffering.

- **Gantt Chart Schedule:**

  A Gantt chart is generated to visually represent the construction schedule. It displays task durations, execution order, and overall project timeline, making the plan easy to interpret.

3.4 Key Outcomes

- Successful autonomous planning from high-level construction goals
- Optimized scheduling with dependency awareness
- Clear visualization of construction timelines
- Demonstration of agent-based AI planning concepts
- Improved understanding of real-world planning problems

4. Conclusion

The Construction Planning Assistant Agent effectively demonstrates the application of Agentic AI concepts in construction planning. The system automates task breakdown, resource validation, scheduling, risk analysis, and visualization, resulting in a structured and optimized execution plan.

Through this project, key concepts such as agent-based design, AI-driven planning, modular system architecture, and practical problem-solving were successfully applied and understood.

5. Future Scope & Enhancements

- Integration with real construction resource and cost APIs
- Persistent storage for project plans
- Support for multi-goal and multi-project planning
- Advanced risk prediction using historical data

- Automated re-planning based on execution feedback
- Enhanced user authentication and access control